

CHEATSHEET · QUICK REFERENCE

Lab 15 — Virtualization with KVM / QEMU

Define a VM with virt-install, inspect bridged/NAT/isolated virtual switches, apply overprovisioning rules and compare cloud delivery models.

Host / guest and virtual disk

```
egrep -c '(vmx|svm)' /proc/cpuinfo
>0 = host CPU supports hardware virt

qemu-img create -f qcow2 vm1.qcow2 5G
Thin-provisioned virtual disk

qemu-img info vm1.qcow2
Inspect the image
```

Define and inspect a VM

```
virt-install --name vm1 --memory 512 --vcpus 1 --disk path=vm1.qcow2 --network network=default --in
Define a VM

virsh list --all
List VMs

virsh net-list --all
List virtual networks

ip addr show virbr0
Inspect the NAT bridge
```

Virtual networking modes

Mode	Guest sees	Use
Bridged	IP on the host LAN	Production VMs
NAT	Private IP, host NATs	Lab / desktop VMs
Isolated	Only other VMs on the vSwitch	Inter-VM lab nets

Overprovisioning + cloud models

Resource / Model	Rule / definition
CPU	4:1 vCPU:pCPU general, 1:1 latency-sensitive
Memory	Never overprovision past physical RAM
Disk	Thin-provision but monitor utilisation
Public	Provider-owned, multi-tenant (AWS/Azure)
Private	Single-tenant (VMware/Proxmox/OpenStack)
Hybrid	Mix spanning both (Azure Arc, Outposts)